(19) Push and hold differential case toward pinion gear and zero dial indicator (Fig. 21).

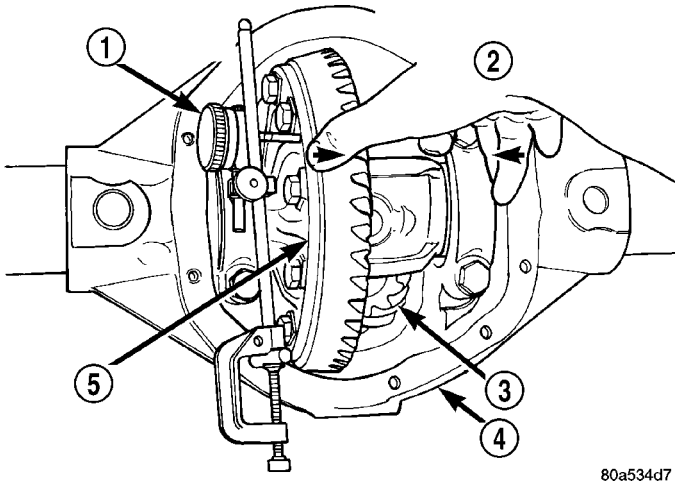


Fig. 21 ZERO DIAL INDICATOR

1 - DIAL INDICATOR FACE
2 - DIFFERENTIAL CASE TO PINION GEAR SIDE
3 - PINION GEAR
4 - DIFFERENTIAL HOUSING
5 - DIFFERENTIAL CASE

(20) Push and hold differential case to ring gear side of the housing and record dial indicator reading (Fig. 22). Subtract 0.05 mm (0.002 in.) from this reading. This is the shim thickness for the ring gear side.

NOTE: This is the shim needed on the ring gear side for proper backlash.

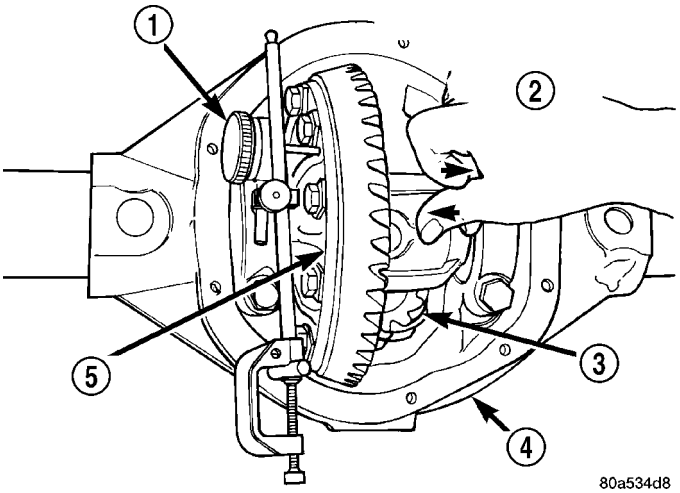


Fig. 22 RECORD DIAL INDICATOR READING

1 - DIAL INDICATOR
2 - DIFFERENTIAL CASE TO RING GEAR SIDE
3 - PINION GEAR
4 - DIFFERENTIAL HOUSING
5 - DIFFERENTIAL CASE

(21) Subtract the backlash shim thickness from the total preload shim thickness. The remainder is the shim thickness required on the pinion side of the housing.

(22) Rotate dial indicator out of the way on pilot stud.

(23) Remove differential case and dummy bearings from the housing.

(24) Install the selected shims onto the differential case hubs.

(25) Install side bearings on differential case hubs with Install C-3716-A and Handle C-4171.

(26) Install bearing cups on differential.

(27) Install Spreader W-129-B and some items from Adapter Set 6987 on the housing and spread open enough to receive differential case.

CAUTION: Do not spread housing over 0.38 mm (0.015 in.). Failure to heed caution may result in damage.

(28) Install differential case into the housing.

(29) Remove spreader from the housing.

(30) Rotate the differential case several times to seat the side bearings.

(31) Position the indicator plunger against a ring gear tooth (Fig. 23).

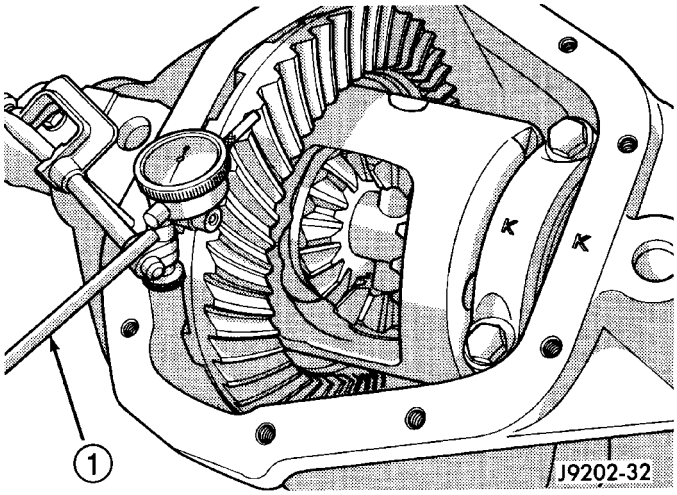


Fig. 23 RING GEAR BACKLASH MEASUREMENT

1 - DIAL INDICATOR

(32) Push and hold ring gear upward while not allowing the pinion gear to rotate.

(33) Zero dial indicator face to pointer.

(34) Push and hold ring gear downward while not allowing the pinion gear to rotate. Dial indicator reading should be between 0.12 mm - 0.20 mm (0.005 in. - 0.008 in.). If backlash is not within specifications transfer the necessary amount of shim thickness from one side of the housing to the other (Fig. 24).